

Shaik Mohammed Feroz

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Summary

Embedded Software Engineer with 4+ years of experience designing and testing firmware for embedded Linux devices and ARM-based microcontrollers. Proficient in C, C++, and Python scripting, with a proven track record developing high-performance communication protocols and automation tests. Collaborates effectively across technical teams and with customers to meet design requirements and drive project success.

Education

Fairfield University | *Master's, computer software engineering*

May 2025

Sathyabama Institute of Science and Technology | *BE, Electronic and Communication Engineering*

Jun 2021

Technical Skills

- **Programming & Development:** C, C++, Python, Embedded C, Object-Oriented Programming, XML, JSON, OS Programming, device driver development, Microcontroller Communication Drivers
- **Embedded Platforms & Tools:** Linux OS, ARM Cortex Microcontrollers, RTOS, Wireshark, cmake, makefile, Embedded Linux, Electronics Engineering
- **Firmware & Testing:** Unit Testing (Google Test Framework), Automation Testing, Integration Testing, Firmware Development
- **Protocols & Communication:** TCP/IP, UDP, Open-Thread, I2C, SPI, UART, IoT Communication Frameworks
- **Version Control & Collaboration:** Git, Bitbucket, Confluence, Jira, Artifactory
- **Software Development Lifecycle (SDLC):** Agile Development, UML Modeling, Software Integration, CI/CD Test Automation

Work Experience

ASML (Wilton, CT) | *Embedded Software Engineer*

June 2024 – Till date

- Developed embedded applications using C/C++ and validated the code with Python automation scripts.
- Customized C++ executables using make file rules to run on x86 and ppce6500 hardware targets.
- Developed interfaces to communicate between Reticle Handler (RH) and other components in an EUV lithography system.
- Designed a new process for handling special wafers (reticles) in EXE 5000, a variant of EUV lithography machine.
- Implemented client-server applications as part of the design process for handling special reticles in EXE 5000.
- Compiled software requirements and design decisions into individual work items on Polarion ALM.
- Integrated code changes on VSCode, compiled using Gradle, and performed code reviews.
- Validated low level driver code using Python scripts and Google Test framework.
- Utilized Valgrind, SonarQube for memory debugging tool to check for memory leaks and overflows.
- Built custom patches and tested them on all variants of EUV based TWINSKAN machine, a proprietary product of ASML.
- Worked in Agile Scrum environment and used JIRA to track and analyze tasks.
- Mentored a group of Interns to onboard by training them on the SDLC at ASML.
- Aligned with Scrum Master regularly to compile sprint planning report and discuss future for EXE 5000.

Aurestech (Fairfield, CT) | *Embedded Software Engineer*

Oct 2023 - May 2024

- Engineered and optimized embedded firmware for 5G-enabled drones, ensuring high-speed data transfer and reliable real-time communication.
- Collaborated with cross-functional teams to implement embedded system software and document technical findings in alignment with project milestones.
- Coordinated development activities and applied integration testing practices to validate system performance on critical hardware platforms.
- Contributed to the extension of the internal test framework and performed performance regression testing during firmware upgrades.

Lutron Electronics (Mysore, India) | *Software Embedded Engineer*

Jan 2022 - Jun 2023

- Developed embedded features for home automation systems in C++ on Linux OS.
- Built and extended automation test suites using Python and Google Test Framework.
- Conducted protocol-level debugging using Wireshark and contributed to a Live Channel Change feature enabling seamless switching without rebooting.
- Collaborated with hardware teams on microcontroller interfacing and performance validation.
- Designed test plans and executed manual and automated tests across integration layers, ensuring end-to-end reliability

- Developed and tested microcontroller drivers in Embedded C and C++, refining I2C, SPI, UART, and GPIO communication for seamless hardware interaction.
- Collaborated with senior engineers to develop low-level firmware for ARM-based microcontrollers, ensuring compliance with project specifications.
- Documented software design and testing processes, contributing to knowledge sharing and team efficiency.

Academic Projects

Big Cat Vocalization Detection System

- Developed Python-based backend for automated detection/classification of animal vocalizations.
- Created a web-based application with Django and JS for visualizing and validating audio detections.
- Integrated MySQL for efficient data storage and search functionalities.
- Deployed the application on Microsoft Azure with secure and scalable architecture.

Visual Aesthetic Influence on Audiences preferences

- Designed and implemented a machine learning model to analyze the correlation between visual aesthetics and IMDb movie ratings.
- Extracted key visual features such as color composition, brightness, saturation, and texture from movie stills using OpenCV.
- Performed correlation analysis to identify visual trends influencing audience reception, producing insights for filmmakers to optimize cinematographic styles.
- Presented findings through comprehensive visualizations, including scatterplots, histograms, and heatmaps to highlight key insights.

RTOS-Based Task Scheduler Simulation

- Designed a modular task scheduler simulation in C/C++ using FreeRTOS to emulate real-world embedded task coordination.
- Implemented multiple task threads with varying priorities to simulate concurrent execution and verified thread safety with semaphore/mutex protection.
- Simulated CPU scheduling and context switching scenarios, validating preemptive behavior and resource sharing.
- Integrated message queues and inter-task signals for effective communication and control flow in real-time operations.

UART Communication Simulator

- Created a UART protocol simulation using Python (PySerial) and C++ to emulate asynchronous serial communication between virtual devices.
- Built encoding/decoding logic with packet structure, framing characters, CRC checks, and retransmission logic for noise simulation.
- Developed a terminal-like interface with real-time command execution, including verbose error flags, signal tracing, and loopback test scenarios.
- Evaluated buffer overflow scenarios, transmission delays, and byte-misalignment cases to model realistic embedded communication challenges.

Embedded Linux-Based File System Emulator

- Built a file system emulator using QEMU, Linux kernel, and C for embedded Linux devices.
- Implemented key features such as wear leveling, efficient read/write operations, and robust file recovery mechanisms.
- Enhanced understanding of file system architecture, Linux internals, and device driver development concepts.

Key Achievements

- **Spot Award** | Received a Spot Award within 3 months for outstanding client feedback on new feature development and deployment and Enhanced data accuracy through optimized protocol implementation and rigorous testing.